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JEE MAIN · CHEMISTRY · CHAPTER-WISE PYQS

Ionic Equilibrium

JEE Main Chemistry · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	112 questions
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MCQ (Single Correct Answer) 112 questions

Q1. JEE Main 2026 (Online) 6th April Evening Shift

Open solution

Given is a concentrated solution of a weak electrolyte A_xB_y of concentration ' c ' and dissociation constant ' K '. The degree of dissociation is given by :

(A) $\left[\mathrm{K} \times \mathrm{c}^{x+y-1} x^x y^y\right]^{x+y}$

(B) $\left(\frac{\mathrm{K}}{\mathrm{c}^{x+y-1} x^x y^y}\right)^{\frac{1}{x+y}}$

(C) $\left(\frac{\mathrm{c}^{x+y-1} x^x y^y}{\mathrm{K}}\right)^{x+y}$

(D) $\left(\frac{\mathrm{c}^{x+y-1} x^x y^y}{\mathrm{K}}\right)^{\frac{1}{x+y}}$

Q2. JEE Main 2026 (Online) 5th April Morning Shift

Open solution

$\mathrm{M}_3\ \mathrm{A}_2$ is a sparingly soluble salt of molar mass $y\ \mathrm{g\ mol}^{-1}$ and solubility $x\ \mathrm{g\ L}^{-1}$. The ratio of the molar concentration of the anion $\left(\mathrm{A}^{3-}\right)$ to the solubility product of the salt is

(A) $\frac{1}{54} \cdot \frac{y^4}{x^4}$

(B) $\frac{y^5}{108x^4}$

(C) $108 \cdot \frac{x^5}{y^5}$

(D) $\frac{1}{108} \frac{y^4}{x^4}$

Q3. JEE Main 2026 (Online) 5th April Morning Shift

Open solution

Arrange the following resultant mixtures in increasing order of their pH values

A. $10\ \mathrm{mL} 0.2 \mathrm{M} \mathrm{Ca(OH)_2} + 25\ \mathrm{mL} 0.1 \mathrm{M} \mathrm{HCl}$

B. $10\ \mathrm{mL} 0.01 \mathrm{M} \mathrm{H_2SO_4} + 10\ \mathrm{mL} 0.01 \mathrm{M} \mathrm{Ca(OH)_2}$

C. $10\ \mathrm{mL} 0.1 \mathrm{M} \mathrm{H_2SO_4} + 10\ \mathrm{mL} 0.1 \mathrm{M} \mathrm{KOH}$

Choose the correct answer from the options given below :

(A) $\mathrm{B} < \mathrm{C} < \mathrm{A}$

(B) $\mathrm{C} < \mathrm{A} < \mathrm{B}$

(C) $\mathrm{C} < \mathrm{B} < \mathrm{A}$

(D) $\mathrm{A} < \mathrm{C} < \mathrm{B}$

Q4. JEE Main 2026 (Online) 5th April Morning Shift

Open solution

Given below are two statements :

Statement I : Sodium dichromate and potassium dichromate are classified as primary standards in titrimetric analysis.

Statement II : Phenolphthalein is a weak base, therefore it dissociates in acidic medium.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are ture

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

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Q5. JEE Main 2026 (Online) 4th April Evening Shift

20 mL of a solution of acetic acid required 28.4 mL of 0.1 M NaOH for its neutralization. A solution (X) was prepared by mixing 20 mL of the above acetic acid and 14.2 mL of 0.1 M NaOH solution. What is the pH of the solution (X) ? $(\mathrm{pK_a}$ value of acetic acid is 4.75).

Open solution

(A) 7.0 (B) 4.75 (C) 3.5 (D) 4.82

Q6. JEE Main 2026 (Online) 2nd April Evening Shift

The first and second ionization constants of a weak dibasic acid H_2A are 8.1×10^{-8} and 1.0×10^{-13} respectively. 0.1 mol of H_2A was dissolved in 1 L of 0.1 M HCl solution. The concentration of HA^- in the resultant solution is:

Open solution

(A) $0.1\ \mathrm{M}$ (B) $9.53 \times 10^{-6}\ \mathrm{M}$ (C) $8.1 \times 10^{-8}\ \mathrm{M}$ (D) $1.0 \times 10^{-13}\ \mathrm{M}$

Q7. JEE Main 2026 (Online) 2nd April Evening Shift

At 25°C, 20.0 mL of 0.2 M weak monoprotic acid HX is titrated against 0.2 M NaOH. The pH of the solution (a) at the start of the titration (when NaOH has not been added) and (b) when 10 mL of NaOH is added respectively, are :

Open solution

Given :
$K_a = 5 \times 10^{-4}$
$\mathrm{p}K_a = 3.3$
$\alpha \ll 1$

(A)

(a)	(b)
0.7	2.0

(B)

(a)	(b)
2.0	3.3

(C)

(a)	(b)
1.1	2.2

(D)

(a)	(b)
3.0	2.2

Q8. JEE Main 2026 (Online) 2nd April Morning Shift

The solubility product constants of $\mathrm{Ag_2CrO_4}$ and AgBr are $32x$ and $4y$ respectively at 298 K.

The value of $\left(\frac{\text{molarity of } \mathrm{Ag_2CrO_4}}{\text{molarity of } \mathrm{AgBr}}\right)$ can be expressed as :

Open solution

(A) $\frac{2\sqrt[3]{x}}{y}$ (B) $2\sqrt{\frac{x}{y}}$ (C) $\sqrt{\frac{x}{y}}$ (D) $\frac{\sqrt[3]{x}}{\sqrt{y}}$

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Q9. JEE Main 2026 (Online) 28th January Morning Shift

Open solution

Consider a weak base ' B ' of $\mathrm{pK_b} = 5.699$. ' x ' mL of 0.02 M HCl and ' y ' mL of 0.02 M weak base ' B ' are mixed to make 100 mL of a buffer of pH 9 at $25^\circ\mathrm{C}$. The values of ' x ' and ' y ' respectively are :

[Given : $\log 2 = 0.3010, \log 3 = 0.4771, \log 5 = 0.699$]

(A) x y
42.7 57.3

(B) x y
14.3 85.7

(C) x y
85.7 14.3

(D) x y
11.1 88.9

Q10. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

Which of the following mixture gives a buffer solution with $\mathrm{pH} = 9.25$?

Given : $\mathrm{pK_b}\ (\mathrm{NH_4OH}) = 4.75$

(A) $0.5\mathrm{M}\mathrm{NH_4OH}(0.2\ \mathrm{L}) + 0.2\mathrm{M}\mathrm{HCl}(0.5\ \mathrm{L})$

(B) $0.2\mathrm{M}\mathrm{NH_4OH}(0.5\ \mathrm{L}) + 0.1\mathrm{M}\mathrm{HCl}(0.5\ \mathrm{L})$

(C) $0.2\mathrm{M}\mathrm{NH_4OH}(0.4\ \mathrm{L}) + 0.1\mathrm{M}\mathrm{HCl}(1\ \mathrm{L})$

(D) $0.4\mathrm{M}\mathrm{NH_4OH}(1\ \mathrm{L}) + 0.1\mathrm{M}\mathrm{HCl}(1\ \mathrm{L})$

Q11. JEE Main 2025 (Online) 7th April Morning Shift

Open solution

An aqueous solution of HCl with pH 1.0 is diluted by adding equal volume of water (ignoring dissociation of water). The pH of HCl solution would

(Given $\log 2 = 0.30$)

(A) increase to 1.3 (B) reduce to 0.5 (C) increase to 2 (D) remain same

Q12. JEE Main 2025 (Online) 2nd April Morning Shift

Open solution

If equal volumes of AB_2 and XY (both are salts) aqueous solutions are mixed, which of the following combination will give a precipitate of $\mathrm{AY_2}$ at 300 K ? (Given $\mathrm{K_{sp}}$ (at 300 K) for $\mathrm{AY_2} = 5.2 \times 10^{-7}$)

(A) $2.0 \times 10^{-4}\mathrm{M}\mathrm{AB_2}, 0.8 \times 10^{-3}\mathrm{M}\mathrm{XY}$

(B) $2.0 \times 10^{-2}\mathrm{M}\mathrm{AB_2}, 2.0 \times 10^{-2}\mathrm{M}\mathrm{XY}$

(C) $1.5 \times 10^{-4}\mathrm{M}\mathrm{AB_2}, 1.5 \times 10^{-3}\mathrm{M}\mathrm{XY}$

(D) $3.6 \times 10^{-3}\mathrm{M}\mathrm{AB_2}, 5.0 \times 10^{-4}\mathrm{M}\mathrm{XY}$

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Q13. JEE Main 2025 (Online) 28th January Evening Shift

Open solution

Arrange the following in increasing order of solubility product :

$Ca(OH)_2, AgBr, PbS, HgS$

(A) $PbS < HgS < Ca(OH)_2 < AgBr$

(B) $HgS < AgBr < PbS < Ca(OH)_2$

(C) $HgS < PbS < AgBr < Ca(OH)_2$

(D) $Ca(OH)_2 < AgBr < HgS < PbS$

Q14. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

A weak acid HA has degree of dissociation x . Which option gives the correct expression of (pH - pK_a)?

(A) $\log\left(\frac{1-x}{x}\right)$

(B) 0

(C) $\log(1+2x)$

(D) $\log\left(\frac{x}{1-x}\right)$

Q15. JEE Main 2025 (Online) 24th January Morning Shift

Open solution

K_{sp} for $Cr(OH)_3$ is 1.6×10^{-30}. What is the molar solubility of this salt in water?

(A) $\sqrt[5]{1.8 \times 10^{-30}}$

(B) $\frac{1.8\times10^{-30}}{27}$

(C) $\sqrt[4]{\frac{1.6\times10^{-30}}{27}}$

(D) $\sqrt[2]{1.6 \times 10^{-30}}$

Q16. JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

pH of water is 7 at $25^\circ C$. If water is heated to $80^\circ C$., it's pH will :

(A) Decrease

(B) Remains the same

(C) Increase

(D) H^+concentration increases, OH^-concentration decreases

Q17. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

Which of the following happens when NH_4OH is added gradually to the solution containing 1 M A^{2+} and $1MB^{3+}$ ions?

Given : $K_{sp}\ [A(OH)_2] = 9 \times 10^{-10}$ and $K_{sp}\ [B(OH)_3] = 27 \times 10^{-18}$ at 298 K.

(A) $A(OH)_2$ will precipitate before $B(OH)_3$

(B) $A(OH)_2$ and $B(OH)_3$ will precipitate together

(C) Both $A(OH)_2$ and $B(OH)_3$ do not show precipitation with NH_4OH

(D) $B(OH)_3$ will precipitate before $A(OH)_2$

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Q18. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

The molar solubility(s) of zirconium phosphate with molecular formula $\left(Zr^{4+}\right)_3\left(PO_4^{3-}\right)_4$ is given by relation :

(A) $\left(\frac{K_{sp}}{5348}\right)^{\frac{1}{6}}$

(B) $\left(\frac{K_{sp}}{8435}\right)^{\frac{1}{7}}$

(C) $\left(\frac{K_{sp}}{6912}\right)^{\frac{1}{7}}$

(D) $\left(\frac{K_{sp}}{9612}\right)^{\frac{1}{3}}$

Q19. JEE Main 2024 (Online) 9th April Evening Shift

Open solution

For a sparingly soluble salt AB_2, the equilibrium concentrations of A^{2+} ions and B^- ions are $1.2 \times 10^{-4}M$ and $0.24 \times 10^{-3}M$, respectively. The solubility product of AB_2 is :

(A) 0.069×10^{-12} (B) 0.276×10^{-12} (C) 6.91×10^{-12} (D) 27.65×10^{-12}

Q20. JEE Main 2024 (Online) 8th April Evening Shift

Open solution

Given below are two statements :

Statement (I) : A Buffer solution is the mixture of a salt and an acid or a base mixed in any particular quantities

Statement (II) : Blood is a naturally occurring buffer solution whose pH is maintained by $H_2CO_3/HCO_3^{\ominus}$ concentrations.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

Q21. JEE Main 2024 (Online) 8th April Evening Shift

Open solution

The equilibrium $Cr_2O_7^{2-} \rightleftharpoons 2CrO_4^{2-}$ is shifted to the right in :

(A) a weakly acidic medium

(B) a basic medium

(C) a neutral medium

(D) an acidic medium

Q22. JEE Main 2024 (Online) 1st February Evening Shift

Open solution

Solubility of calcium phosphate (molecular mass, M) in water is W_g per $100\ mL$ at $25^\circ C$. Its solubility product at $25^\circ C$ will be approximately.

(A) $10^7\left(\frac{W}{M}\right)^3$

(B) $10^3\left(\frac{W}{M}\right)^5$

(C) $10^7\left(\frac{W}{M}\right)^5$

(D) $10^5\left(\frac{W}{M}\right)^5$

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Q23. JEE Main 2024 (Online) 27th January Morning Shift

Open solution

Given below are two statements :

Statement (I) : Aqueous solution of ammonium carbonate is basic.

Statement (II) : Acidic/basic nature of salt solution of a salt of weak acid and weak base depends on K_a and K_b value of acid and the base forming it.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are correct

(B) Statement I is correct but Statement II is incorrect

(C) Both Statement I and Statement II are incorrect

(D) Statement I is incorrect but Statement II is correct

Q24. JEE Main 2023 (Online) 15th April Morning Shift

Open solution

Which of the following statement(s) is/are correct?

(A) The pH of $1 \times 10^{-8}\ \mathrm{M}\ \mathrm{HCl}$ solution is 8 .

(B) The conjugate base of $\mathrm{H_2PO_4^-}$ is $\mathrm{HPO_4^{2-}}$.

(C) $\mathrm{K_w}$ increases with increase in temperature.

(D) When a solution of a weak monoprotic acid is titrated against a strong base at half neutralisation point, $\mathrm{pH} = \frac{1}{2}\mathrm{pK_a}$

Choose the correct answer from the options given below:

(A) $(\mathrm{A}), (\mathrm{B}), (\mathrm{C})$

(B) (B), (C)

(C) (B), (C), (D)

(D) (A), (D)

Q25. JEE Main 2023 (Online) 11th April Morning Shift

Open solution

$25\ \mathrm{mL}$ of silver nitrate solution (1M) is added dropwise to $25\ \mathrm{mL}$ of potassium iodide $(1.05\mathrm{M})$ solution. The ion(s) present in very small quantity in the solution is/are :

(A) $\mathrm{I^-}$ only

(B) $\mathrm{K^+}$ only

(C) $\mathrm{Ag^+}$ and $\mathrm{I^-}$ both

(D) $\mathrm{NO_3^-}$ only

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Q26. JEE Main 2023 (Online) 31st January Evening Shift

The incorrect statement for the use of indicators in acid-base titration is :

(A) Phenolphthalein is a suitable indicator for a weak acid vs strong base titration.

(B) Methyl orange may be used for a weak acid vs weak base titration.

(C) Methyl orange is a suitable indicator for a strong acid vs weak base titration.

(D) Phenolphthalein may be used for a strong acid vs strong base titration.

Open solution

Q27. JEE Main 2023 (Online) 25th January Evening Shift

When the hydrogen ion concentration [H^+] changes by a factor of 1000, the value of pH of the solution _____

(A) decreases by 2 units

(B) increases by 2 units

(C) decreases by 3 units

(D) increases by 1000 units

Open solution

Q28. JEE Main 2022 (Online) 29th July Evening Shift

$200\ \text{mL}$ of $0.01\ \text{M}\ \text{HCl}$ is mixed with $400\ \text{mL}$ of $0.01\ \text{M}\ \text{H}_2\text{SO}_4$. The pH of the mixture is _____.

Given: $\log 2 = 0.30, \log 3 = 0.48, \log 5 = 0.70, \log 7 = 0.84, \log 11 = 1.04$

(A) 1.14

(B) 1.78

(C) 2.34

(D) 3.02

Open solution

Q29. JEE Main 2022 (Online) 29th July Evening Shift

A compound 'X' is a weak acid and it exhibits colour change at pH close to the equivalence point during neutralization of NaOH with CH_3COOH. Compound 'X' exists in ionized form in basic medium. The compound 'X' is

(A) methyl orange

(B) methyl red

(C) phenolphthalein

(D) erichrome Black T

Open solution

Q30. JEE Main 2022 (Online) 28th July Evening Shift

Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Permanganate titrations are not performed in presence of hydrochloric acid.

Reason R : Chlorine is formed as a consequence of oxidation of hydrochloric acid.

In the light of the above statements, choose the correct answer from the options given below

(A) Both A and R are true and R is the correct explanation of A

(B) Both A and R are true but R is NOT the correct explanation of A

(C) A is true but R is false

(D) A is false but R is true

Open solution

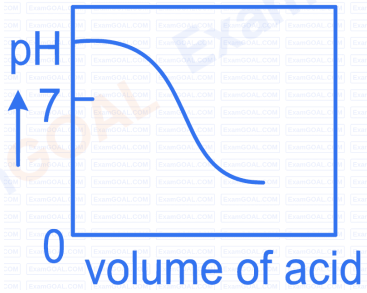
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Q31. JEE Main 2022 (Online) 27th July Evening Shift

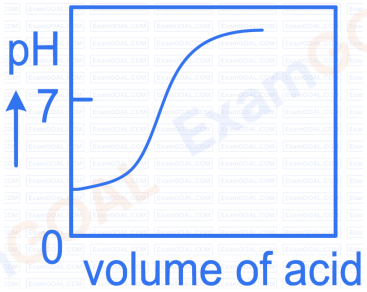
Open solution

The plot of pH-metric titration of weak base NH_4OH vs strong acid HCl looks like :

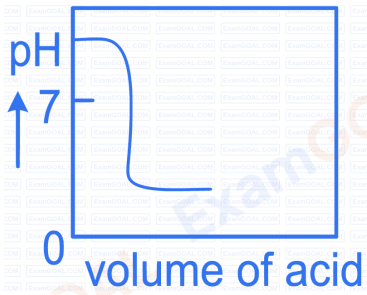
(A)



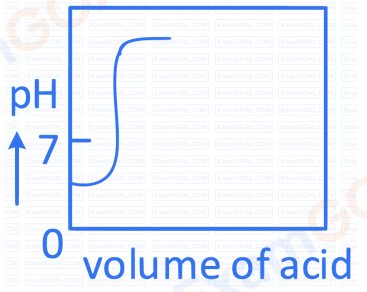
(B)



(C)



(D)



Q32. JEE Main 2022 (Online) 26th July Evening Shift

Open solution

Class XII students were asked to prepare one litre of buffer solution of $pH\ 8.26$ by their Chemistry teacher. The amount of ammonium chloride to be dissolved by the student in $0.2\ M$ ammonia solution to make one litre of the buffer is :

(Given: $pK_b\ (NH_3) = 4.74$

Molar mass of $NH_3 = 17\ g\ mol^{-1}$

Molar mass of $NH_4Cl = 53.5\ g\ mol^{-1}$)

(A) 53.5 g (B) 72.3 g (C) 107.0 g (D) 126.0 g

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Q33. JEE Main 2022 (Online) 25th July Evening Shift

K_{a_1}, K_{a_2} and K_{a_3} are the respective ionization constants for the following reactions (a), (b) and (c).

(a) $H_2C_2O_4 \rightleftharpoons H^+ + HC_2O_4^-$

(b) $HC_2O_4^- \rightleftharpoons H^+ + C_2O_4^{2-}$

(c) $H_2C_2O_4 \rightleftharpoons 2H^+ + C_2O_4^{2-}$

The relationship between K_{a_1}, K_{a_2} and K_{a_3} is given as :

(A) $K_{a_3} = K_{a_1} + K_{a_2}$

(B) $K_{a_3} = K_{a_1} - K_{a_2}$

(C) $K_{a_3} = K_{a_1} / K_{a_2}$

(D) $K_{a_3} = K_{a_1} \times K_{a_2}$

Open solution

Q34. JEE Main 2022 (Online) 25th July Morning Shift

$20\ \text{mL}$ of $0.1\ \text{M}\ NH_4OH$ is mixed with $40\ \text{mL}$ of $0.05 MHCl$. The pH of the mixture is nearest to :

(Given : $K_b\ (NH_4OH) = 1 \times 10^{-5}, \log 2 = 0.30, \log 3 = 0.48, \log 5 = 0.69, \log 7 = 0.84, \log 11 = 1.04$)

(A) 3.2

(B) 4.2

(C) 5.2

(D) 6.2

Open solution

Q35. JEE Main 2022 (Online) 29th June Morning Shift

The solubility of AgCl will be maximum in which of the following?

(A) 0.01 M KCl

(B) 0.01 M HCl

(C) 0.01 M $AgNO_3$

(D) Deionised water

Open solution

Q36. JEE Main 2022 (Online) 28th June Evening Shift

A student needs to prepare a buffer solution of propanoic acid and its sodium salt with pH 4. The ratio of $\frac{[CH_3CH_2COO^-]}{[CH_3CH_2COOH]}$ required to make buffer is _____.

Given : $K_a(CH_3CH_2COOH) = 1.3 \times 10^{-5}$

(A) 0.03

(B) 0.13

(C) 0.23

(D) 0.33

Open solution

Q37. JEE Main 2022 (Online) 25th June Evening Shift

The K_{sp} for bismuth sulphide (Bi_2S_3) is 1.08×10^{-73}. The solubility of Bi_2S_3 in mol L^{-1} at 298 K is :

(A) 1.0×10^{-15}

(B) 2.7×10^{-12}

(C) 3.2×10^{-10}

(D) 4.2×10^{-8}

Open solution

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Q38. JEE Main 2022 (Online) 25th June Evening Shift

Open solution

Given below are two statements one is labelled as Assertion A and the other is labelled as Reason R :

Assertion A : The amphoteric nature of water is explained by using Lewis acid/base concept.

Reason R : Water acts as an acid with NH_3 and as a base with H_2S.

In the light of the above statements choose the correct answer from the options given below :

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Q39. JEE Main 2021 (Online) 26th August Morning Shift

Open solution

Given below are two statements.

Statement I : In the titration between strong acid and weak base methyl orange is suitable as an indicator.

Statement II : For titration of acetic acid with NaOH phenolphthalein is not a suitable indicator.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Q40. JEE Main 2021 (Online) 20th July Evening Shift

Open solution

A solution is 0.1 M in Cl^- and 0.001 M in CrO_4^{2-}. Solid $AgNO_3$ is gradually added to it. Assuming that the addition does not change in volume and $K_{sp}(AgCl) = 1.7 \times 10^{-10}\ M^2$ and $K_{sp}(Ag_2CrO_4) = 1.9 \times 10^{-12}\ M^3$.

Select correct statement from the following :

(A) AgCl precipitates first because its K_{sp} is high.

(B) Ag_2CrO_4 precipitates first as its K_{sp} is low.

(C) Ag_2CrO_4 precipitates first because the amount of Ag^+ needed is low.

(D) AgCl will precipitate first as the amount of Ag^+ needed to precipitate is low.

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Q41. JEE Main 2021 (Online) 18th March Morning Shift

Open solution

Given below are two statements : One is labelled as Assertion A and the other labelled as reason R

Assertion A : During the boiling of water having temporary hardness, $Mg(HCO_3)_2$ is converted to $MgCO_3$.

Reason R : The solubility product of $Mg(OH)_2$ is greater than that of $MgCO_3$.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both A and R are true but R is not the correct explanation of A

(B) A is true but R is false

(C) Both A and R are true and R is the correct explanation of A

(D) A is false but R is true

Q42. JEE Main 2021 (Online) 17th March Morning Shift

Open solution

Which of the following compound CANNOT act as a Lewis base?

(A) SF_4 (B) NF_3 (C) ClF_3 (D) PCl_5

Q43. JEE Main 2021 (Online) 25th February Evening Shift

Open solution

The solubility of $Ca(OH)_2$ in water is :

[Given : The solubility product of $Ca(OH)_2$ in water = 5.5×10^{-6}]

(A) 1.11×10^{-6} (B) 1.11×10^{-2} (C) 1.77×10^{-6} (D) 1.77×10^{-2}

Q44. JEE Main 2021 (Online) 25th February Morning Shift

Open solution

The solubility of AgCN in a buffer solution of pH = 3 is x. The value of x is : [Assume : No cyano complex is formed; $K_{sp}(AgCN) = 2.2 \times 10^{-16}$ and $K_a(HCN) = 6.2 \times 10^{-10}$]

(A) 1.9×10^{-5} (B) 1.6×10^{-6} (C) 2.2×10^{-16} (D) 0.625×10^{-6}

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Q45. JEE Main 2020 (Online) 6th September Morning Slot

Open solution

Arrange the following solutions in the decreasing order of pOH :

(A) 0.01 M HCl
(B) 0.01 M NaOH
(C) 0.01 M CH_3COONa
(D) 0.01 M NaCl

(A) (B) > (C) > (D) > (A)

(B) (A) > (D) > (C) > (B)

(C) (A) > (C) > (D) > (B)

(D) (B) > (D) > (C) > (A)

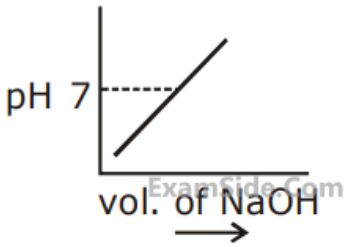
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Q46. JEE Main 2020 (Online) 3rd September Evening Slot

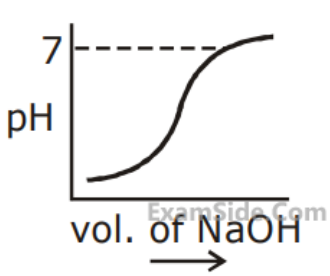
100 mL of 0.1 M HCl is taken in a beaker and to it 100 mL of 0.1 M NaOH is added in steps of 2 mL and the pH is continuously measured. Which of the following graphs correctly depicts the change in pH?

Open solution

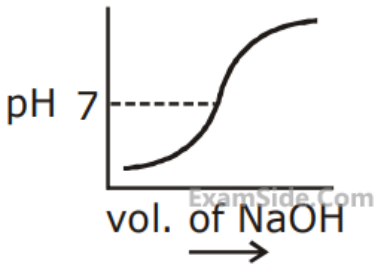
(A)



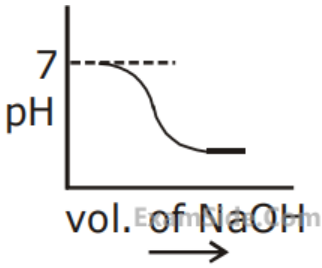
(B)



(C)



(D)



Q47. JEE Main 2020 (Online) 3rd September Morning Slot

An acidic buffer is obtained on mixing :

Open solution

(A) 100 mL of 0.1 M HCl and 200 mL of 0.1 M CH_3COONa

(B) 100 mL of 0.1 M HCl and 200 mL of 0.1 M NaCl

(C) 100 mL of 0.1 M CH_3COOH and 100 mL of 0.1 M NaOH

(D) 100 mL of 0.1 M CH_3COOH and 200 mL of 0.1 M NaOH

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Q48. JEE Main 2020 (Online) 2nd September Morning Slot

Open solution

For the following **Assertion** and **Reason**, the correct option is

Assertion (A): When Cu (II) and sulphide ions are mixed, they react together extremely quickly to give a solid.

Reason (R): The equilibrium constant of
$Cu^{2+}(aq) + S^{2-}(aq) \rightleftharpoons CuS(s)$ is high because the solubility product is low.

(A) (A) is false and (R) is true.

(B) Both (A) and (R) are true but (R) is not the explanation for (A).

(C) Both (A) and (R) are true and (R) is the explanation for (A).

(D) Both (A) and (R) are false.

Q49. JEE Main 2020 (Online) 9th January Evening Slot

Open solution

The solubility product of $Cr(OH)_3$ at 298 K is 6.0×10^{-31}. The concentration of hydroxide ions in a saturated solution of $Cr(OH)_3$ will be :

(A) $(2.22 \times 10^{-31})^{1/4}$

(B) $(4.86 \times 10^{-29})^{1/4}$

(C) $(18 \times 10^{-31})^{1/4}$

(D) $(18 \times 10^{-31})^{1/2}$

Q50. JEE Main 2020 (Online) 9th January Morning Slot

Open solution

The K_{sp} for the following dissociation is 1.6×10^{-5}

$$PbCl_{2(s)} \leftrightharpoons Pb^{2+}_{(aq)} + 2Cl^{-}_{(aq)}$$

Which of the following choices is correct for a mixture of 300 mL 0.134 M $Pb(NO_3)_2$ and 100 mL 0.4 M NaCl ?

(A) $Q > K_{sp}$

(B) Not enough data provided

(C) $Q < K_{sp}$

(D) $Q = K_{sp}$

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Q51. JEE Main 2020 (Online) 8th January Evening Slot

Open solution

For the following Assertion and Reason, the correct option is :

Assertion : The pH of water increases with increase in temperature.

Reason : The dissociation of water into H^+ and OH^- is an exothermic reaction.

(A) Both assertion and reason are false.

(B) Both assertion and reason are true, but the reason is not the correct explanation for the assertion.

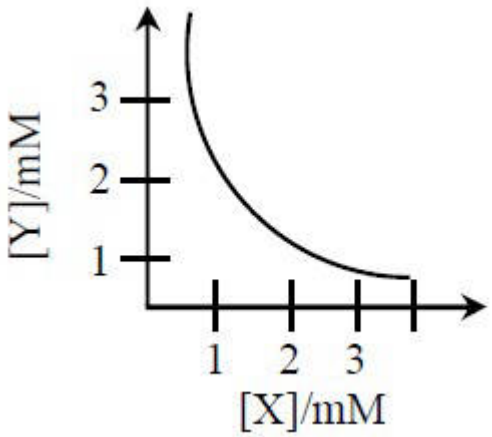
(C) Both assertion and reason are true, and the reason is the correct explanation for the assertion.

(D) Assertion is not true, but reason is true.

Q52. JEE Main 2020 (Online) 8th January Morning Slot

Open solution

The stoichiometry and solubility product of a salt with the solubility curve given below is, respectively :



(A) XY, 2×10^{-6} M^3 (B) XY_2, 1×10^{-9} M^3 (C) XY_2, 4×10^{-9} M^3 (D) X_2Y, 2×10^{-9} M^3

Q53. JEE Main 2020 (Online) 8th January Morning Slot

Open solution

The strength of an aqueous NaOH solution is most accurately determined by titrating :
(Note : consider that an appropriate indicator is used)

(A) Aq. NaOH in a pipette and aqueous oxalic acid in a burette

(B) Aq. NaOH in a burette and aqueous oxalic acid in a conical flask

(C) Aq. NaOH in a volumetric flask and concentrated H_2SO_4 in a conical flask

(D) Aq. NaOH in a burette and concentrated H_2SO_4 in a conical flask

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Q54. JEE Main 2019 (Online) 12th April Evening Slot

The decreasing order of electrical conductivity of the following aqueous solutions is :

0.1 M Formic acid (A),

0.1 M Acetic acid (B),

0.1 M Benzoic acid (C)

(A) A > B > C **(B)** C > B > A **(C)** A > C > B **(D)** C > A > B

Open solution

Q55. JEE Main 2019 (Online) 12th April Evening Slot

The molar solubility of $Cd(OH)_2$ is 1.84×10^{-5} M in water. The expected solubility of $Cd(OH)_2$ in a buffer solution of pH = 12 is :

(A) 2.49×10^{-10} M **(B)** 1.84×10^{-9} M

(C) 6.23×10^{-11} M **(D)** $\frac{2.49}{1.84} \times 10^{-9} M$

Open solution

Q56. JEE Main 2019 (Online) 12th April Morning Slot

What is the molar solubility of $Al(OH)_3$ in 0.2 M NaOH solution ? Given that, solubility product of $Al(OH)_3 = 2.4 \times 10^{-24}$:

(A) 3×10^{-22} **(B)** 3×10^{-19} **(C)** 12×10^{-21} **(D)** 12×10^{-22}

Open solution

Q57. JEE Main 2019 (Online) 10th April Evening Slot

The pH of a 0.02 M NH_4Cl solution will be :

[given K_b $(NH_4OH) = 10^{-5}$ and log 2 = 0.301]

(A) 2.56 **(B)** 5.35 **(C)** 4.35 **(D)** 4.65

Open solution

Q58. JEE Main 2019 (Online) 10th April Morning Slot

Consider the following statements

(a) The pH of a mixture containing 400 mL of 0.1 M H_2SO_4 and 400 mL of 0.1 M NaOH will be approximately 1.3

(b) Ionic product of water is temperature dependent.

(c) A monobasic acid with Ka = 10^{-5} has pH = 5. The degree of dissociation of this acid is 50 %.

(d) The Le Chatelier's principle is not applicable to common-ion effect.

The correct statements are :

(A) (a) and (b) **(B)** (a), (b) and (c) **(C)** (a), (b) and (d) **(D)** (b) and (c)

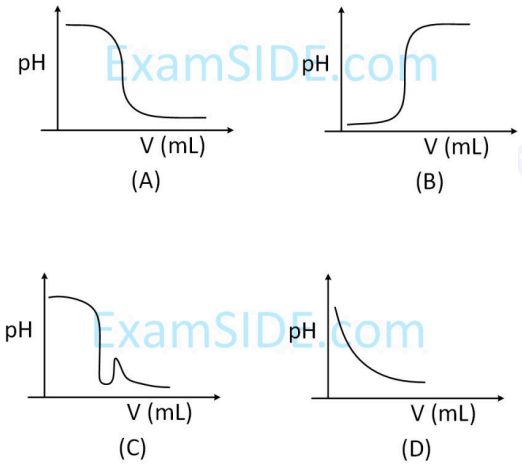
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Q59. JEE Main 2019 (Online) 9th April Evening Slot

In an acid-base titration, 0.1 M HCl solution was added to the NaOH solution of unknown strength. Which of the following correctly shows the change of pH of the titration mixture in this experiment?

Open solution



(A) (C) (B) (A) (C) (B) (D) (D)

Q60. JEE Main 2019 (Online) 8th April Morning Slot

If solublity product of $Zr_3(PO_4)_4$ is denoted by K_{sp} and its molar solubility is denoted by S, then which of the following relation between S and K_{sp} is correct ?

Open solution

(A) $S = \left(\frac{K_{sp}}{929}\right)^{1/9}$ (B) $S = \left(\frac{K_{sp}}{6912}\right)^{1/7}$

(C) $S = \left(\frac{K_{sp}}{144}\right)^{1/6}$ (D) $S = \left(\frac{K_{sp}}{216}\right)^{1/7}$

Q61. JEE Main 2019 (Online) 12th January Evening Slot

If K_{sp} of Ag_2CO_3 is 8×10^{-12}, the molar solubility of Ag_2CO_3 in 0.1 M $AgNO_3$ is -

Open solution

(A) 8×10^{-12} M (B) 8×10^{-10} M (C) 8×10^{-13} M (D) 8×10^{-11} M

Q62. JEE Main 2019 (Online) 10th January Morning Slot

A mixture of 100 m mol of $Ca(OH)_2$ and 2 g of sodium sulphate was dissolved in water and the volume was made up to 100 mL. The mass of calcium sulphate formed and the concentration of OH^- in resulting solution, respectively, are : (Molar mass of Ca $(OH)_2$, Na_2SO_4 and $CaSO_4$ are 74, 143 and 136 g mol^{-1} , respectively; K_{sp} of $Ca(OH)_2$ is 5.5×10^{-6})

Open solution

(A) 13.6g, 0.28 mol L^{-1} (B) 13.6g, 0.14 mol L^{-1}

(C) 1.9g, 0.28 mol L^{-1} (D) 1.9g, 0.14 mol L^{-1}

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Q63. JEE Main 2019 (Online) 9th January Evening Slot

The pH of rain water, is approximately :

(A) 5.6 (B) 7.5 (C) 7.0 (D) 6.5

Open solution

Q64. JEE Main 2019 (Online) 9th January Morning Slot

20 mL of 0.1 M H_2SO_4 solution is added to 30 mL of 0.2 M NH_4OH solution. The pH of the resultant mixture is : [pk_b of NH_4OH = 4.7].

(A) 5.2 (B) 9.0 (C) 5.0 (D) 9.4

Open solution

Q65. JEE Main 2018 (Offline)

An aqueous solution contains an unknown concentration of Ba^{2+}. When 50 mL of a 1 M solution of Na_2SO_4 is added, $BaSO_4$ just begins to precipitate. The final volume is 500 mL. The solubility product of $BaSO_4$ is 1×10^{-10}. What is the original concentration of Ba^{2+}?

(A) 1.0×10^{-10} M (B) 5×10^{-9} M (C) 2×10^{-9} M (D) 1.1×10^{-9} M

Open solution

Q66. JEE Main 2018 (Offline)

An aqueous solution contains 0.10 M H_2S and 0.20 M HCl. If the equilibrium constants for the formation of HS^- from H_2S is 1.0×10^{-7} and that of S^{2-} from HS^- ions is 1.2×10^{-13} then the concentration of S^{2-} ions in aqueous solution is :

(A) 5×10^{-19} (B) 5×10^{-8} (C) 3×10^{-20} (D) 6×10^{-21}

Open solution

Q67. JEE Main 2018 (Offline)

Which of the following are Lewis acids?

(A) BCl_3 and $AlCl_3$ (B) PH_3 and BCl_3 (C) $AlCl_3$ and $SiCl_4$ (D) PH_3 and $SiCl_4$

Open solution

Q68. JEE Main 2018 (Offline)

Which of the following salts is the most basic in aqueous solution?

(A) $Pb(CH_3COO)_2$ (B) $Al(CN)_3$ (C) CH_3COOK (D) $FeCl_3$

Open solution

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Q69. JEE Main 2018 (Offline)

An alkali is titrated against an acid with methyl orange as indicator, which of the following is a correct combination?

Open solution

(A)

Base	Acid	End point
Weak	Strong	Colourless to pink

(B)

Base	Acid	End point
Strong	Strong	Pinkish red to yellow

(C)

Base	Acid	End point
Weak	Strong	Yellow to pinkish red

(D)

Base	Acid	End point
Strong	Strong	Pink to colourless

Q70. JEE Main 2018 (Online) 15th April Evening Slot

Following four solutions are prepared by mixing different volumes of NaOH and HCl of different concentrations, pH of which one of them will be equal to 1 ?

Open solution

(A) 100 mL $\frac{M}{10}$ HCl + 100 mL $\frac{M}{10}$ NaOH

(B) 75 mL $\frac{M}{5}$ HCl + 25 mL $\frac{M}{5}$ NaOH

(C) 60 mL $\frac{M}{10}$ HCl + 40 mL $\frac{M}{10}$ NaOH

(D) 55 mL $\frac{M}{10}$ HCl + 45 mL $\frac{M}{10}$ NaOH

Q71. JEE Main 2018 (Online) 15th April Morning Slot

The minimum volume of water required to dissolve 0.1 g lead (II) chloride to get a saturated solution (K_{sp} of $PbCl_2$ = 3.2 $\times$ 10^{-8}, atomic mass of Pb = 207 u) is :

Open solution

(A) 0.36 L

(B) 17.98 L

(C) 0.18 L

(D) 1.798 L

Q72. JEE Main 2018 (Online) 15th April Morning Slot

Which of the following is a Lewis acid?

Open solution

(A) PH_3

(B) $B(CH_3)_3$

(C) NaH

(D) NF_3

Q73. JEE Main 2017 (Online) 9th April Morning Slot

50 mL of 0.2 M ammonia solution is treated with 25 mL of 0.2 M HCl. If pK_b of ammonia solution is 4.75, the pH of the mixture will be :

Open solution

(A) 3.75

(B) 4.75

(C) 8.25

(D) 9.25

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Q74. JEE Main 2017 (Online) 8th April Morning Slot

Addition of sodium hydroxide solution to a weak acid (HA) results in a buffer of pH 6. If ionisation constant of HA is 10^{-5}, the ratio of salt to acid concentration in the buffer solution will be :

(A) 4 : 5 (B) 1 : 10 (C) 10 : 1 (D) 5 : 4

Open solution

Q75. JEE Main 2017 (Offline)

pK_a of a weak acid (HA) and pK_b of a weak base (BOH) are 3.2 and 3.4, respectively. The pH of their salt (AB) solution is :

(A) 6.9 (B) 7.0 (C) 1.0 (D) 7.2

Open solution

Q76. JEE Main 2014 (Online) 19th April Morning Slot

Zirconium phosphate $[Zr_3(PO_4)_4]$ dissociates into three zirconium cations of charge + 4 and four phosphate anions of charge $-$ 3. If molar solubility of zirconium phosphate is denoted by S and its solubility product by K_{sp} then which of the following relationship between S and K_{sp} is correct ?

(A) $S = \{K_{sp}/(6912)^{1/7}\}$ (B) $S = \{K_{sp}/144\}^{1/7}$

(C) $S = (K_{sp}/6912)^{1/7}$ (D) $S = \{K_{sp}/6912\}^{7}$

Open solution

Q77. JEE Main 2014 (Online) 12th April Morning Slot

The conjugate base of hydrazoic acid is :

(A) N^{-3} (B) N_3^- (C) N_2^- (D) HN_3^-

Open solution

Q78. JEE Main 2014 (Online) 11th April Morning Slot

Assuming that the degree of hydrolysis is small, the pH of 0.1 M solution of sodium acetate ($K_a = 1.0 \times 10^{-5}$) will be :

(A) 5.0 (B) 6.0 (C) 8.0 (D) 9.0

Open solution

Q79. JEE Main 2014 (Online) 11th April Morning Slot

In some solutions, the concentration of H_3O^+ remains constant even when small amounts of strong acid or strong base are added to them. These solutions are known as :

(A) Ideal solutions (B) Colloidal solutions

(C) True solutions (D) Buffer solutions

Open solution

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Q80. JEE Main 2013 (Online) 25th April Morning Slot

What would be the pH of a solution obtained by mixing 5 g of acetic acid and 7.5 g of sodium acetate and making the volume equal to 500 mL? ($K_a = 1.75 \times 10^{-5}$, pK_a = 4.76)

(A) pH = 4.70

(B) pH < 4.70

(C) pH of solution will be equal to pH of acetic acid

(D) 4.76 < pH < 5.0

Open solution

Q81. JEE Main 2013 (Online) 25th April Morning Slot

Which one of the following arrangements represents the correct order of the proton affinity of the given species :

(A) $I^- < F^- < HS^- < NH_2^-$

(B) $HS^- < NH_2^- < F^- < I^-$

(C) $F^- < I^- < NH_2^- < HS^-$

(D) $NH_2^- < HS^- < I^- < F^-$

Open solution

Q82. JEE Main 2013 (Online) 23th April Morning Slot

What is the pH of a 10^{-4} M OH^- solution at 330 K, if K_w at 330 K is $10^{-13.6}$?

(A) 4 (B) 9.0 (C) 10 (D) 9.6

Open solution

Q83. JEE Main 2013 (Online) 22th April Morning Slot

Which one of the following arrangements represents the correct order of solubilities of sparingly soluble salts Hg_2Cl_2, $Cr_2(SO_4)_3$, $BaSO_4$ and $CrCl_3$ respectively?

(A) $BaSO_4 > Hg_2Cl_2 > Cr_2(SO_4)_3 > CrCl_3$

(B) $BaSO_4 > Hg_2Cl_2 > CrCl_3 > Cr_2(SO_4)_3$

(C) $BaSO_4 > CrCl_3 > Hg_2Cl_2 > Cr_2(SO_4)_3$

(D) $Hg_2Cl_2 > BaSO_4 > CrCl_3 > Cr_2(SO_4)_3$

Open solution

Q84. JEE Main 2013 (Online) 22th April Morning Slot

NaOH is a strong base. What will be pH of 5.0×10^{-2} M NaOH solution? (log 2 = 0.3)

(A) 14.00 (B) 13.70 (C) 13.00 (D) 12.70

Open solution

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Q85. JEE Main 2013 (Online) 22th April Morning Slot

Open solution

Values of dissociation constant, K_a are given as follows :

Acid	K_a
HCN	6.2×10^{-10}
HF	7.2×10^{-4}
HNO_2	4.0×10^{-4}

Correct order of increasing base strength of CN^-, F^- and NO_2^- will be :

(A) $F^- < CN^- < NO_2^-$ (B) $NO_2^- < CN^- < F^-$

(C) $F^- < NO_2^- < CN^-$ (D) $NO_2^- < F^- < CN^-$

Q86. JEE Main 2013 (Online) 9th April Morning Slot

Open solution

Solid $Ba(NO_3)_2$ is gradually dissolved in a 1.0×10^{-4} M Na_2CO_3 solution. At which concentration of Ba^{2+}, precipitate of $BaCO_3$ begins to form? (K_{sp} for $BaCO_3 = 5.1 \times 10^{-9}$)

(A) 5.1×10^{-5} M (B) 7.1×10^{-8} M (C) 4.1×10^{-5} M (D) 8.1×10^{-7} M

Q87. JEE Main 2013 (Offline)

Open solution

How many litres of water must be added to 1 litre of an aqueous solution of HCl with a pH of 1 to create an aqueous solution with pH of 2?

(A) 0.1 L (B) 0.9 L (C) 2.0 L (D) 9.0 L

Q88. AIEEE 2012 (Online) 26th May

Open solution

The solubility of PbI_2 at $25^\circ C$ is $0.7\ g\ L^{-1}$. The solubility product of PbI_2 at this temperature is (molar mass of $PbI_2 = 461.2\ g\ mol^{-1}$)

(A) 1.40×10^{-9} (B) 0.14×10^{-9} (C) 140×10^{-9} (D) 14.0×10^{-9}

Q89. AIEEE 2012 (Online) 19th May

Open solution

If K_{sp} of CaF_2 at $25^\circ C$ is 1.7×10^{-10}, the combination amongst the following which gives a precipitate of CaF_2 is

(A) $1 \times 10^{-2}\ M\ Ca^{2+}$ and $1 \times 10^{-3}\ M\ F^-$

(B) $1 \times 10^{-4}\ M\ Ca^{2+}$ and $1 \times 10^{-4}\ M\ F^-$

(C) $1 \times 10^{-2}\ M\ Ca^{2+}$ and $1 \times 10^{-5}\ M\ F^-$

(D) $1 \times 10^{-3}\ M\ Ca^{2+}$ and $1 \times 10^{-5}\ M\ F^-$

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Q90. AIEEE 2012 (Online) 12th May

Open solution

Given (i) $HCN(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + CN^-(aq) \quad K_a = 6.2 \times 10^{-10}$
(ii) $CN^-(aq) + H_2O(l) \rightleftharpoons HCN(aq) + OH^-(aq) \quad K_b = 1.6 \times 10^{-5}$.
These equilibria show the following order of the relative base strength,

(A) $OH^- > H_2O > CN^-$
(B) $OH^- > CN^- > H_2O$
(C) $H_2O > CN^- > OH^-$
(D) $CN^- > H_2O > OH^-$

Q91. AIEEE 2012 (Online) 7th May

Open solution

The solubility (in mol L^{-1}) of AgCl ($K_{sp} = 1.0 \times 10^{-10}$) in a 0.1 M KCl solution will be

(A) 1.0×10^{-9}
(B) 1.0×10^{-10}
(C) 1.0×10^{-5}
(D) 1.0×10^{-11}

Q92. AIEEE 2012

Open solution

The pH of a 0.1 molar solution of the acid HQ is 3. The value of the ionization constant, K_a of this acid is :

(A) 3×10^{-1}
(B) 1×10^{-3}
(C) 1×10^{-5}
(D) 1×10^{-7}

Q93. AIEEE 2010

Open solution

Three reactions involving $H_2PO_4^-$ are given below :

(i) $H_3PO_4 + H_2O \rightarrow H_3O^+ + H_2PO_4^-$

(ii) $H_2PO_4^- + H_2O \rightarrow HPO_4^{2-} + H_3O^+$

(iii) $H_2PO_4^- + OH^- \rightarrow H_3PO_4 + O^{2-}$

In which of the above does $H_2PO_4^-$ act as an acid?

(A) (ii) only
(B) (i) and (ii)
(C) (iii) only
(D) (i) only

Q94. AIEEE 2010

Open solution

In aqueous solution the ionization constants for carbonic acid are
$K_1 = 4.2 \times 10^{-7}$ and $K_2 = 4.8 \times 10^{-11}$
Select the correct statement for a saturated 0.034 M solution of the carbonic acid.

(A) The concentration of CO_3^{2-} is 0.034 M.
(B) The concentration of CO_3^{2-} is greater than that of HCO_3^-
(C) The concentration of H^+ and HCO_3^- are approximately equal.
(D) The concentration of H^+ is double that of CO_3^{2-}.

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Q95. AIEEE 2010

Solubility product of silver bromide is 5.0×10^{-13}. The quantity of potassium bromide (molar mass taken as 120g of mol^{-1}) to be added to 1 litre of 0.05 M solution of silver nitrate to start the precipitation of AgBr is :

Open solution

(A) 1.2×10^{-10} g (B) 1.2×10^{-9} g (C) 6.2×10^{-5} g (D) 5.0×10^{-8} g

Q96. AIEEE 2010

At 25°C, the solubility product of $Mg(OH)_2$ is 1.0×10^{-11}. At which pH, will Mg^{2+} ions start precipitating in the form of $Mg(OH)_2$ from a solution of 0.001 M Mg^{2+} ions?

Open solution

(A) 9 (B) 10 (C) 11 (D) 8

Q97. AIEEE 2009

Solid $Ba(NO_3)_2$ is gradually dissolved in a 1.0×10^{-4} M Na_2CO_3 solution. At what concentration of Ba^{2+} will a precipitate begin to form ?
(K_{sp} for $BaCO_3 = 5.1 \times 10^{-9}$)

Open solution

(A) 5.1×10^{-5} M (B) 8.1×10^{-8} M (C) 8.1×10^{-7} M (D) 4.1×10^{-5} M

Q98. AIEEE 2008

Four species are listed below
i. HCO_3^-
ii. H_3O^+
iii. HSO_4^-
iv. HSO_3F
Which one of the following is the correct sequence of their acid strength?

Open solution

(A) iv < ii < iii < i (B) ii < iii < i < iv
(C) i < iii < ii < iv (D) iii < i < iv < ii

Q99. AIEEE 2008

The pK_a of a weak acid, HA, is 4.80. The pK_b of a weak base, BOH, is 4.78. The pH of an aqueous solution of the corresponding salt, BA, will be

Open solution

(A) 9.58 (B) 4.79 (C) 7.01 (D) 9.22

Q100. AIEEE 2007

The first and second dissociation constants of an acid H_2A are 1.0×10^{-5} and 5.0×10^{-10} respectively. The overall dissociation constant of the acid will be :

Open solution

(A) 5.0×10^{-5} (B) 5.0×10^{15} (C) 5.0×10^{-15} (D) 5.0×10^{5}

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Q101. AIEEE 2007

The pK_a of a weak acid (HA) is 4.5. The pOH of an aqueous buffered solution of HA in which 50% of the acid is ionized is :

(A) 7.0 (B) 4.5 (C) 2.5 (D) 9.5

Open solution

Q102. AIEEE 2007

In a saturated solution of the sparingly soluble strong electrolyte $AgIO_3$ (Molecular mass = 283) the equilibrium which sets in is

$AgIO_{3(s)} \rightleftharpoons Ag^+(aq) + IO_3^-$

If the solubility product constant K_{sp} of $AgIO_3$ at a given temperature is 1.0×10^{-8}, what is the mass of $AgIO_3$ contained in 100 ml of its saturated solution?

(A) 28.3×10^{-2} g (B) 2.83×10^{-3} g (C) 1.0×10^{-7} g (D) 1.0×10^{-4} g

Open solution

Q103. AIEEE 2005

The solubility product of a salt having general formula MX_2, in water is: 4×10^{-12}. The concentration of M^{2+} ions in the aqueous solution of the salt is :

(A) 2.0×10^{-6} M (B) 4.0×10^{-10} M (C) 1.0×10^{-4} M (D) 1.6×10^{-4} M

Open solution

Q104. AIEEE 2005

Hydrogen ion concentration in mol / L in a solution of pH = 5.4 will be :

(A) 3.98×10^{-6} (B) 3.68×10^{-6} (C) 3.88×10^{6} (D) 3.98×10^{8}

Open solution

Q105. AIEEE 2005

What is the conjugate base of OH^-?

(A) O_2 (B) H_2O (C) O^- (D) O^{-2}

Open solution

Q106. AIEEE 2004

The conjugate base of $H_2PO_4^-$ is :

(A) PO_4^{3-} (B) HPO_4^{2-} (C) H_3PO_4 (D) P_2O_5

Open solution

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Q107. AIEEE 2004

The molar solubility (in mol L^{-1}) of a sparingly soluble salt MX_4 is "s". The corresponding solubility product is K_{sp}. 's' is given in terms of K_{sp} by the relation :

(A) $s = (256\ K_{sp})^{1/5}$

(B) $s = (128\ K_{sp})^{1/4}$

(C) $s = (K_{sp} / 128)^{1/4}$

(D) $s = (K_{sp} / 256)^{1/5}$

Open solution

Q108. AIEEE 2003

Which one of the following statements is not true?

(A) pH + pOH = 14 for all aqueous solutions

(B) The pH of 1×10^{-8} M HCl is 8

(C) 96,500 coulombs of electricity when passed through a $CuSO_4$ solution deposits 1 gram equivalent of copper at the cathode

(D) The conjugate base of $H_2PO_4^-$ is HPO_4^{2-}

Open solution

Q109. AIEEE 2003

The solubility in water of a sparingly soluble salt AB_2 is 1.0×10^{-5} mol L^{-1}. Its solubility product number will be :

(A) 4×10^{-10}

(B) 1×10^{-15}

(C) 1×10^{-10}

(D) 4×10^{-15}

Open solution

Q110. AIEEE 2002

1 M NaCl and 1 M HCl are present in an aqueous solution. The solution is

(A) not a buffer solution with pH < 7

(B) not a buffer solution with pH > 7

(C) a buffer solution with pH < 7

(D) a buffer solution with pH > 7

Open solution

Q111. AIEEE 2002

Species acting as both Bronsted acid and base is :

(A) $(HSO_4)^{-1}$

(B) Na_2CO_3

(C) NH_3

(D) OH^{-1}

Open solution

Q112. AIEEE 2002

Let the solubility of an aqueous solution of $Mg(OH)_2$ be x then its K_{sp} is :

(A) $4x^3$

(B) $108x^5$

(C) $27x^4$

(D) $9x$

Open solution

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Numerical 46 questions

Q113. JEE Main 2026 (Online) 4th April Morning Shift

The pH of a solution obtained by mixing 5 mL of $0.1 \mathrm{M} \mathrm{NH}_4\mathrm{OH}$ solution with 250 mL of $0.1 \mathrm{M} \mathrm{NH}_4\mathrm{Cl}$ solution is ____ $\times 10^{-2}$.(Nearest integer) Given: $\mathrm{pK}_{\mathrm{b}}\left(\mathrm{NH}_4\mathrm{OH}\right)=4.74$

$\log 2=0.30$
$\log 3=0.48$
$\log 5=0.70$

Open solution

Q114. JEE Main 2026 (Online) 28th January Morning Shift

Consider the dissociation equilibrium of the following weak acid

$\mathrm{HA} \rightleftharpoons \mathrm{H}^{+}(\mathrm{aq})+\mathrm{A}^{-}(\mathrm{aq})$

If the pKa of the acid is 4 , then the pH of 10 mM HA solution is ____ .(Nearest integer)

[Given: The degree of dissociation can be neglected with respect to unity]

Open solution

Q115. JEE Main 2026 (Online) 24th January Morning Shift

Consider two Group IV metal ions X^{2+} and Y^{2+}.

A solution containing $0.01 \mathrm{M} \mathrm{X}^{2+}$ and $0.01 \mathrm{M} \mathrm{Y}^{2+}$ is saturated with $\mathrm{H}_2\mathrm{S}$. The pH at which the metal sulphide YS will form as a precipitate is ____ . (Nearest integer)

(Given: $\mathrm{K}_{\mathrm{sp}}(\mathrm{XS})=1 \times 10^{-22}$ at $25^{\circ} \mathrm{C}, \mathrm{K}_{\mathrm{sp}}(\mathrm{YS})=4 \times 10^{-16}$ at $25^{\circ} \mathrm{C},\left[\mathrm{H}_2\mathrm{S}\right]=0.1 \mathrm{M}$ in solution, $\mathrm{K}_{a 1} \times \mathrm{K}_{a 2}\left(\mathrm{H}_2\mathrm{S}\right)=1.0 \times 10^{-21}, \log 2=0.30, \log 3=0.48, \log 5=0.70$)

Open solution

Q116. JEE Main 2026 (Online) 21st January Evening Shift

The first and second ionization constants of H_2X are 2.5×10^{-8} and 1.0×10^{-13} respectively.

The concentration of X^{2-} in $0.1\ \mathrm{M}$ H_2X solution is _____ $\times 10^{-15}\ \mathrm{M}$. (Nearest Integer)

Open solution

Q117. JEE Main 2025 (Online) 7th April Evening Shift

One litre buffer solution was prepared by adding 0.10 mol each of NH_3 and $\mathrm{NH}_4\mathrm{Cl}$ in deionised water. The change in pH on addition of 0.05 mol of HCl to the above solution is _____ $\times 10^{-2}$.

(Nearest integer)

Given : pK_{b} of $\mathrm{NH}_3=4.745$ and $\log_{10} 3=0.477$

Open solution

Q118. JEE Main 2025 (Online) 4th April Evening Shift

$x \mathrm{mg}$ of $\mathrm{Mg(OH)_2}$(molar mass $=58$) is required to be dissolved in 1.0 L of water to produce a pH of 10.0 at 298 K . The value of x is _____ mg. (Nearest integer)

(Given : $\mathrm{Mg(OH)_2}$ is assumed to dissociate completely in $\mathrm{H_2O}$)

Open solution

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Q119. JEE Main 2025 (Online) 4th April Morning Shift

The pH of a 0.01 M weak acid HX $\left(\mathrm{K}_a = 4 \times 10^{-10}\right)$ is found to be 5 . Now the acid solution is diluted with excess of water so that the pH of the solution changes to 6 . The new concentration of the diluted weak acid is given as $x \times 10^{-4}\mathrm{M}$. The value of x is _____ (nearest integer)

Open solution

Q120. JEE Main 2025 (Online) 23rd January Morning Shift

If 1 mM solution of ethylamine produces $\mathrm{pH} = 9$, then the ionization constant $\left(\mathrm{K_b}\right)$ of ethylamine is 10^{-x}. The value of x is _____ (nearest integer).

[The degree of ionization of ethylamine can be neglected with respect to unity.]

Open solution

Q121. JEE Main 2024 (Online) 6th April Morning Shift

Consider the dissociation of the weak acid HX as given below

$\mathrm{HX(aq)} \rightleftharpoons \mathrm{H^+(aq) + X^-(aq), Ka = 1.2 \times 10^{-5}}$

[$\mathrm{K_a}$: dissociation constant]

The osmotic pressure of $0.03\mathrm{M}$ aqueous solution of HX at $300\ \mathrm{K}$ is _____ $\times 10^{-2}$ bar (nearest integer).

[Given : $\mathrm{R} = 0.083\ \mathrm{L\ bar\ mol^{-1}\ K^{-1}}$]

Open solution

Q122. JEE Main 2024 (Online) 1st February Morning Shift

$\mathrm{K_a}$ for $\mathrm{CH_3COOH}$ is 1.8×10^{-5} and $\mathrm{K_b}$ for $\mathrm{NH_4OH}$ is 1.8×10^{-5}. The pH of ammonium acetate solution will be _____.

Open solution

Q123. JEE Main 2024 (Online) 30th January Evening Shift

The pH of an aqueous solution containing $1\mathrm{M}$ benzoic acid $\left(\mathrm{pK_a} = 4.20\right)$ and $1\mathrm{M}$ sodium benzoate is 4.5. The volume of benzoic acid solution in $300\ \mathrm{mL}$ of this buffer solution is _____ mL. (given : $\log 2 = 0.3$)

Open solution

Q124. JEE Main 2024 (Online) 30th January Morning Shift

The pH at which $\mathrm{Mg(OH)_2}$ $\left[\mathrm{K_{sp}} = 1 \times 10^{-11}\right]$ begins to precipitate from a solution containing $0.10\ \mathrm{M}\ \mathrm{Mg^{2+}}$ ions is _____.

Open solution

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Q125. JEE Main 2023 (Online) 13th April Evening Shift

20 mL of $0.1\ \mathrm{M\ NaOH}$ is added to $50\ \mathrm{mL}$ of $0.1\ \mathrm{M}$ acetic acid solution. The pH of the resulting solution is _____ $\times 10^{-2}$ (Nearest integer)

Given : $\mathrm{pKa\,(CH_3COOH)} = 4.76$

$\log 2 = 0.30$

$\log 3 = 0.48$

Open solution

Q126. JEE Main 2023 (Online) 13th April Morning Shift

$25.0\ \mathrm{mL}$ of $0.050\ \mathrm{M\ Ba(NO_3)_2}$ is mixed with $25.0\ \mathrm{mL}$ of $0.020\ \mathrm{M\ NaF}$. $\mathrm{K_{Sp}}$ of $\mathrm{BaF_2}$ is 0.5×10^{-6} at $298\ \mathrm{K}$. The ratio of $\left[\mathrm{Ba^{2+}}\right]\left[\mathrm{F^-}\right]^2$ and $\mathrm{K_{sp}}$ is _____.

(Nearest integer)

Open solution

Q127. JEE Main 2023 (Online) 12th April Morning Shift

An analyst wants to convert $1\ \mathrm{L\ HCl}$ of $\mathrm{pH} = 1$ to a solution of HCl of $\mathrm{pH}\ 2$. The volume of water needed to do this dilution is _____ mL. (Nearest integer)

Open solution

Q128. JEE Main 2023 (Online) 8th April Evening Shift

The solubility product of $\mathrm{BaSO_4}$ is 1×10^{-10} at $298\ \mathrm{K}$. The solubility of $\mathrm{BaSO_4}$ in $0.1\ \mathrm{M\ K_2SO_4(aq)}$ solution is _____ $\times 10^{-9}\ \mathrm{g\ L^{-1}}$ (nearest integer).

Given: Molar mass of $\mathrm{BaSO_4}$ is $233\ \mathrm{g\ mol^{-1}}$

Open solution

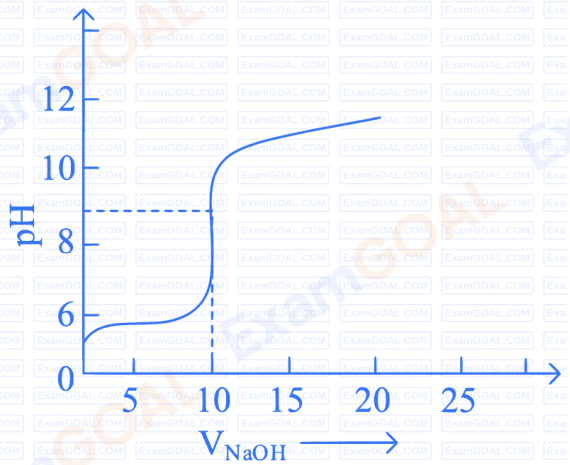
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Q129. JEE Main 2023 (Online) 8th April Morning Shift

Open solution

The titration curve of weak acid vs. strong base (with phenolphthalein as indicator) is shown below. The $K_{phenolphthalein} = 4 \times 10^{-10}$.

Given: $\log 2 = 0.3$



The number of following statement/s which is/are correct about phenolphthalein is _____

A. It can be used as an indicator for the titration of weak acid with weak base.

B. It begins to change colour at $pH = 8.4$

C. It is a weak organic base

D. It is colourless in acidic medium

Q130. JEE Main 2023 (Online) 31st January Evening Shift

Open solution

At $298\ K$, the solubility of silver chloride in water is $1.434 \times 10^{-3}\ g\ L^{-1}$. The value of $-\log K_{sp}$ for silver chloride is _____.

(Given mass of Ag is $107.9\ g\ mol^{-1}$ and mass of Cl is $35.5\ g\ mol^{-1}$)

Q131. JEE Main 2023 (Online) 30th January Morning Shift

Open solution

$600\ mL$ of $0.01\ M\ HCl$ is mixed with $400\ mL$ of $0.01\ M\ H_2SO_4$. The pH of the mixture is _____ $\times 10^{-2}$. (Nearest integer)

[Given $\log 2 = 0.30$

$\log 3 = 0.48$

$\log 5 = 0.69$

$\log 7 = 0.84$

$\log 11 = 1.04$]

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Q132. JEE Main 2023 (Online) 29th January Morning Shift

Millimoles of calcium hydroxide required to produce 100 mL of the aqueous solution of pH 12 is $x \times 10^{-1}$. The value of x is _____ (Nearest integer).

Assume complete dissociation.

Open solution

Q133. JEE Main 2023 (Online) 25th January Morning Shift

A litre of buffer solution contains 0.1 mole of each of NH_3 and NH_4Cl. On the addition of 0.02 mole of HCl by dissolving gaseous HCl, the pH of the solution is found to be _____ $\times 10^{-3}$ (Nearest integer)

[Given : $pK_b(NH_3) = 4.745$

$\log 2 = 0.301$

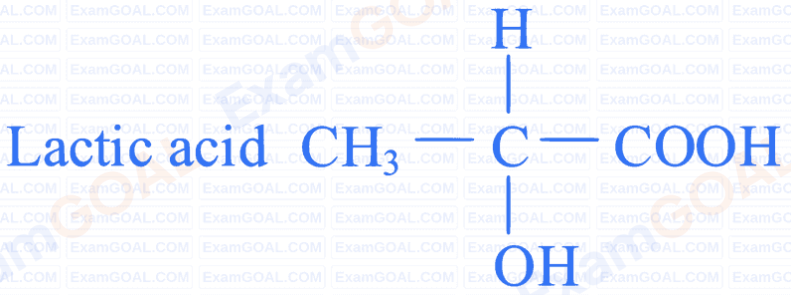
$\log 3 = 0.477$

$T = 298\ K$]

Open solution

Q134. JEE Main 2023 (Online) 24th January Evening Shift

If the pKa of lactic acid is 5, then the pH of 0.005 M calcium lactate solution at 25°C is _____ $\times 10^{-1}$ (Nearest integer)

Lactic acid 

Open solution

Q135. JEE Main 2023 (Online) 24th January Morning Shift

The dissociation constant of acetic acid is $x \times 10^{-5}$. When 25 mL of 0.2 $M\ CH_3COONa$ solution is mixed with 25 mL of 0.02 $M\ CH_3COOH$ solution, the pH of the resultant solution is found to be equal to 5. The value of x is _____

Open solution

Q136. JEE Main 2022 (Online) 30th July Evening Shift

50 mL of 0.1 M CH_3COOH is titrated against 0.1 M NaOH solution. When 10 mL of NaOH is added, the pH of the solution becomes _____ $\times 10^{-1}$. (nearest integer)

Given: pK_a (CH_3COOH) = 4.8, log 2 = 0.3

Open solution

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Q137. JEE Main 2022 (Online) 29th July Morning Shift

If the solubility product of PbS is 8×10^{-28}, then the solubility of PbS in pure water at 298 K is x $\times$ 10^{-16} mol L^{-1}. The value of x is _____. (Nearest Integer)

[Given : $\sqrt{2}$ = 1.41]

Open solution

Q138. JEE Main 2022 (Online) 28th July Morning Shift

K_a for butyric acid (C_3H_7COOH) is 2×10^{-5}. The pH of $0.2\,M$ solution of butyric acid is _____ $\times 10^{-1}$. (Nearest integer)

[Given $\log 2 = 0.30$]

Open solution

Q139. JEE Main 2022 (Online) 27th July Morning Shift

At $310\,K$, the solubility of CaF_2 in water is $2.34 \times 10^{-3}\,g/100\,mL$. The solubility product of CaF_2 is _____ $\times 10^{-8}(\,mol/L)^3$. (Given molar mass : $CaF_2 = 78\,g\,mol^{-1}$)

Open solution

Q140. JEE Main 2022 (Online) 27th July Morning Shift

In the titration of $KMnO_4$ and oxalic acid in acidic medium, the change in oxidation number of carbon at the end point is _____.

Open solution

Q141. JEE Main 2022 (Online) 28th June Morning Shift

The solubility product of a sparingly soluble salt A_2X_3 is 1.1×10^{-23}. If specific conductance of the solution is 3×10^{-5} S m^{-1}, the limiting molar conductivity of the solution is $x \times 10^{-3}$ S m^2 mol^{-1}. The value of x is _____.

Open solution

Q142. JEE Main 2022 (Online) 27th June Evening Shift

pH value of 0.001 M NaOH solution is _____.

Open solution

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Q143. JEE Main 2022 (Online) 26th June Morning Shift

50 mL of 0.1 M CH_3COOH is being titrated against 0.1 M NaOH. When 25 mL of NaOH has been added, the pH of the solution will be _____ $\times 10^{-2}$. (Nearest integer)

(Given : pK_a (CH_3COOH) = 4.76)

log 2 = 0.30

log 3 = 0.48

log 5 = 0.69

log 7 = 0.84

log 11 = 1.04

Open solution

Q144. JEE Main 2021 (Online) 1st September Evening Shift

The molar solubility of $Zn(OH)_2$ in 0.1 M NaOH solution is x $\times 10^{-18}$ M. The value of x is _____ (Nearest integer)

(Given : The solubility product of $Zn(OH)_2$ is 2×10^{-20})

Open solution

Q145. JEE Main 2021 (Online) 31st August Evening Shift

The pH of a solution obtained by mixing 50 mL of 1 M HCl and 30 mL of 1 M NaOH is x $\times 10^{-4}$. The value of x is _____. (Nearest integer) [log 2.5 = 0.3979]

Open solution

Q146. JEE Main 2021 (Online) 31st August Morning Shift

A_3B_2 is a sparingly soluble salt of molar mass M (g mol^{-1}) and solubility x g L^{-1}. The solubility product satisfies $K_{sp} = a\left(\frac{x}{M}\right)^5$. The value of a is _____. (Integer answer)

Open solution

Q147. JEE Main 2021 (Online) 26th August Morning Shift

The OH^- concentration in a mixture of 5.0 mL of 0.0504 M NH_4Cl and 2 mL of 0.0210 M NH_3 solution is x $\times 10^{-6}$ M. The value of x is _____. (Nearest integer)

[Given $K_w = 1 \times 10^{-14}$ and $K_b = 1.8 \times 10^{-5}$]

Open solution

Q148. JEE Main 2021 (Online) 25th July Evening Shift

Assuming that $Ba(OH)_2$ is completely ionised in aqueous solution under the given conditions the concentration of H_3O^+ ions in 0.005 M aqueous solution of $Ba(OH)_2$ at 298 K is _____ $\times 10^{-12}$ mol L^{-1}. (Nearest integer)

Open solution

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Q149. JEE Main 2021 (Online) 18th March Evening Shift

The solubility of $CdSO_4$ in water is 8.0×10^{-4} mol L^{-1}. Its solubility in 0.01 M H_2SO_4 solution is _____ $\times 10^{-6}$ mol L^{-1}. (Round off to the Nearest Integer). (Assume that solubility is much less than 0.01 M)

Open solution

Q150. JEE Main 2021 (Online) 18th March Morning Shift

In order to prepare a buffer solution of pH 5.74, sodium acetate is added to acetic acid. If the concentration of acetic acid in the buffer is 1.0 M, the concentration of sodium acetate in the buffer is _____ M. (Round off to the Nearest Integer). [Given : pKa (acetic acid) = 4.74]

Open solution

Q151. JEE Main 2021 (Online) 16th March Evening Shift

Sulphurous acid (H_2SO_3) has $Ka_1 = 1.7 \times 10^{-2}$ and $Ka_2 = 6.4 \times 10^{-8}$. The pH of 0.588 M H_2SO_3 is _____. (Round off to the Nearest Integer).

Open solution

Q152. JEE Main 2021 (Online) 16th March Morning Shift

Two salts A_2X and MX have the same value of solubility product of 4.0×10^{-12}. The ratio of their molar solubilities i.e. $\frac{S(A_2X)}{S(MX)}$ = _____. (Round off to the Nearest Integer)

Open solution

Q153. JEE Main 2021 (Online) 26th February Evening Shift

The pH of ammonium phosphate solution, if pk_a of phosphoric acid and pk_b of ammonium hydroxide are 5.23 and 4.75 respectively, is _____. (Nearest integer)

Open solution

Q154. JEE Main 2021 (Online) 24th February Evening Shift

The solubility product of PbI_2 is 8.0×10^{-9}. The solubility of lead iodide in 0.1 molar solution of lead nitrate is $x \times 10^{-6}$. mol/L. The value of x is _____. (Rounded off to the nearest integer) [Given $\sqrt{2}$ = 1.41]

Open solution

Q155. JEE Main 2020 (Online) 6th September Evening Slot

If the solubility product of AB_2 is 3.20×10^{-11} M^3, then the solubility of AB_2 in pure water is _____ $\times 10^{-4}$ mol L^{-1}.
[Assuming that neither kind of ion reacts with water]

Open solution

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Q156. JEE Main 2020 (Online) 5th September Morning Slot

A soft drink was bottled with a partial pressure of CO_2 of 3 bar over the liquid at room temperature. The partial pressure of CO_2 over the solution approaches a value of 30 bar when 44 g of CO_2 is dissolved in 1 kg of water at room temperature. The approximate pH of the soft drink is _____ $\times 10^{-1}$.
(First dissociation constant of $H_2CO_3 = 4.0 \times 10^{-7}$; log 2 = 0.3; density of the soft drink = 1 g mL^{-1}) .

Open solution

Q157. JEE Main 2020 (Online) 7th January Evening Slot

3 g of acetic acid is added to 250 mL of 0.1 M HCl and the solution made up to 500 mL. To 20 mL of this solution $\frac{1}{2}$ mL of 5 M NaOH is added. The pH of the solution is _____.

[Given : pKa of acetic acid = 4.75, molar mass of acetic acid = 60 g/mol, log 3 = 0.4771] Neglect any changes in volume.

Open solution

Q158. JEE Main 2020 (Online) 7th January Morning Slot

Two solutions, A and B, each of 100L was made by dissolving 4g of NaOH and 9.8 g of H_2SO_4 in water, respectively. The pH of the resultant solution obtained from mixing 40L of solution A and 10L of solution B is :

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **A**	3. **C**	4. **B**	5. **B**
6. **C**	7. **B**	8. **D**	9. **B**	10. **B**
11. **A**	12. **B**	13. **C**	14. **D**	15. **C**
16. **A**	17. **D**	18. **C**	19. **C**	20. **C**
21. **B**	22. **C**	23. **A**	24. **B**	25. **C**
26. **B**	27. **C**	28. **B**	29. **C**	30. **A**
31. **A**	32. **C**	33. **D**	34. **C**	35. **D**
36. **B**	37. **A**	38. **D**	39. **B**	40. **D**
41. **D**	42. **D**	43. **B**	44. **A**	45. **B**
46. **C**	47. **A**	48. **C**	49. **C**	50. **A**
51. **A**	52. **C**	53. **B**	54. **C**	55. **A**
56. **A**	57. **B**	58. **B**	59. **B**	60. **B**
61. **B**	62. **C**	63. **A**	64. **B**	65. **D**
66. **C**	67. **A**	68. **C**	69. **C**	70. **B**
71. **C**	72. **B**	73. **D**	74. **C**	75. **A**
76. **C**	77. **B**	78. **D**	79. **D**	80. **D**
81. **A**	82. **D**	83. *Scan QR*	84. **D**	85. **C**
86. **A**	87. **D**	88. **D**	89. **A**	90. **B**
91. **A**	92. **C**	93. **A**	94. **C**	95. **B**
96. **B**	97. **A**	98. **C**	99. **C**	100. **C**
101. **D**	102. **B**	103. **C**	104. **A**	105. **D**
106. **B**	107. **D**	108. **B**	109. **D**	110. **A**
111. **A**	112. **A**			

Numerical

113. **756**	114. **3**	115. **4**	116. **100**	117. **48**
118. **3**	119. **25**	120. **7**	121. **76**	122. **7**
123. **100**	124. **9**	125. **458**	126. **5**	127. **9000**
128. **233**	129. **2**	130. **10**	131. **186**	132. **5**
133. **9079**	134. **85**	135. **10**	136. **42**	137. **282**
138. **27**	139. **0**	140. **1**	141. **3**	142. **11**

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143. **476**	144. **2**	145. **6021**	146. **108**	147. **3**
148. **1**	149. **64**	150. **10**	151. **1**	152. **50**
153. **7**	154. **141**	155. **2**	156. **37**	157. **5.22 to 5.24**
158. **10.6**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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